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📄 **Highly Available 3-Tier Web Application on AWS**

🧾 **Project Overview**

This project demonstrates the deployment of a **highly available, scalable, and fault-tolerant web application** using AWS cloud services. The architecture is designed to handle traffic spikes and ensure high availability across multiple Availability Zones.

🏗 **Architecture Description**

The application follows a **3-tier architecture**:

Internet
↓
Application Load Balancer (ALB)
↓
Target Group
↓
EC2 Instances (Auto Scaling Group)
↓
EBS (Application Data)
↓
S3 (Static Assets)

Additional Components:

- Bastion Host (for secure SSH access)
- AWS Backup (for EBS snapshots)

☁ **AWS Services Used**

- Amazon EC2 – Virtual servers for application hosting
- Application Load Balancer – Distributes incoming traffic
- Amazon S3 – Stores static assets

- Auto Scaling – Automatically scales instances
- Amazon EBS – Persistent storage
- AWS Backup – Snapshot automation
- AWS Certificate Manager – HTTPS security

🌐 1. VPC & Networking Setup

Objective:

Create a secure and isolated network.

Steps:

1. Created a **VPC** with CIDR:

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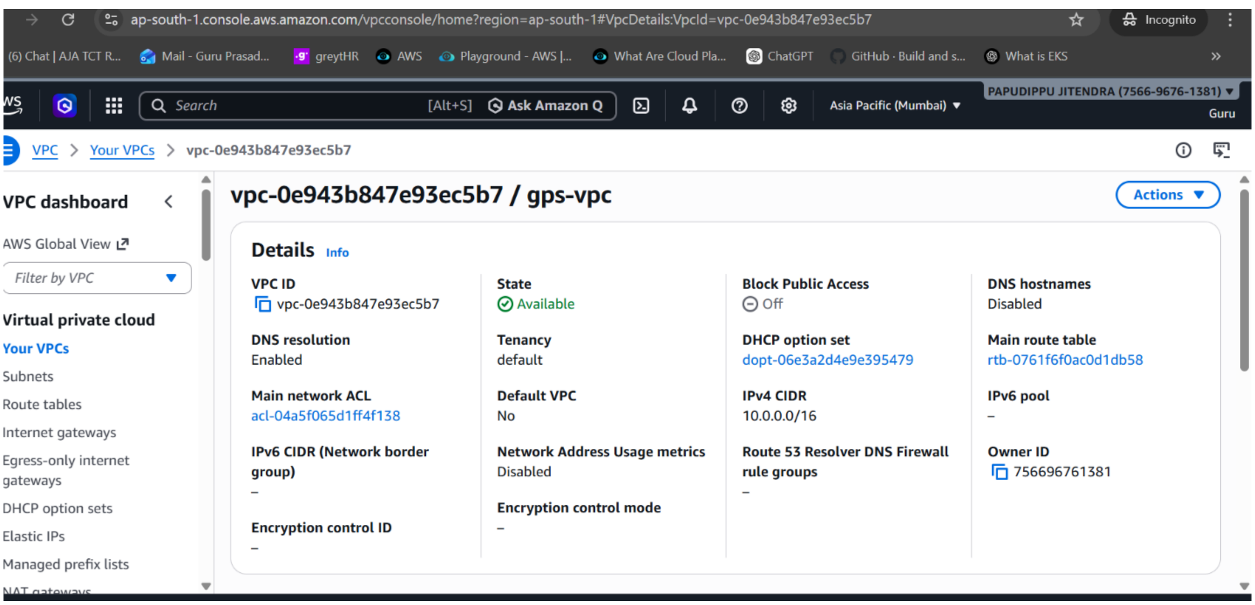
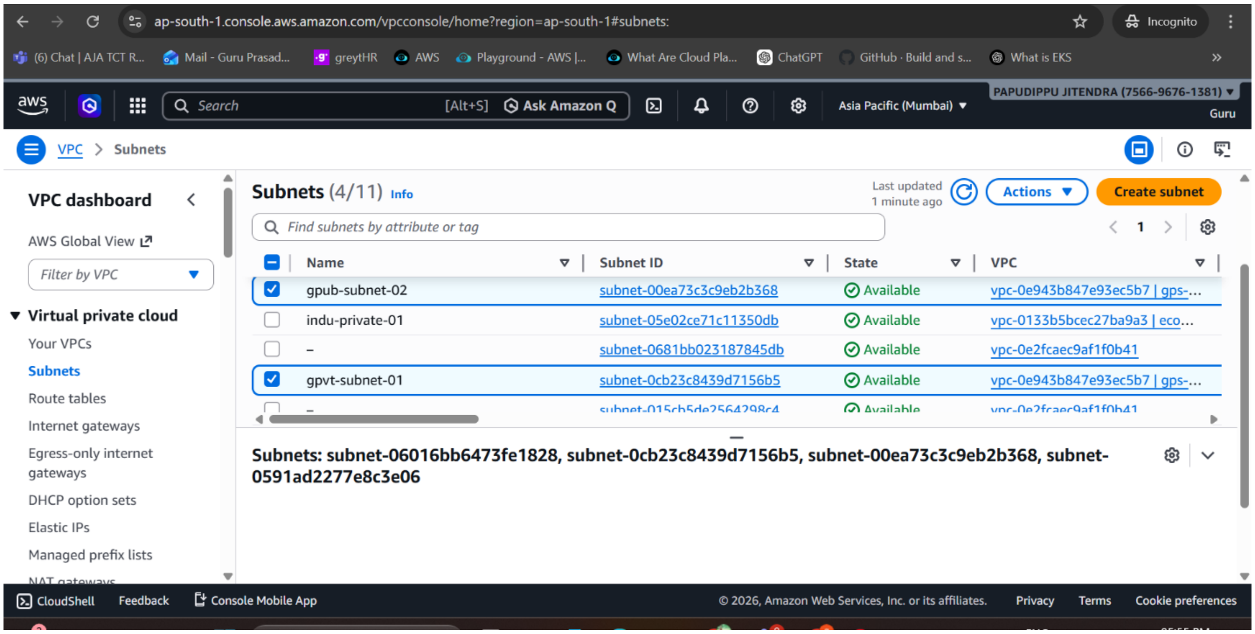


2. Created **Subnets in 2 Availability Zones**:
   - Public Subnet (for ALB & Bastion)
   - Private Subnet (for EC2)

3. 

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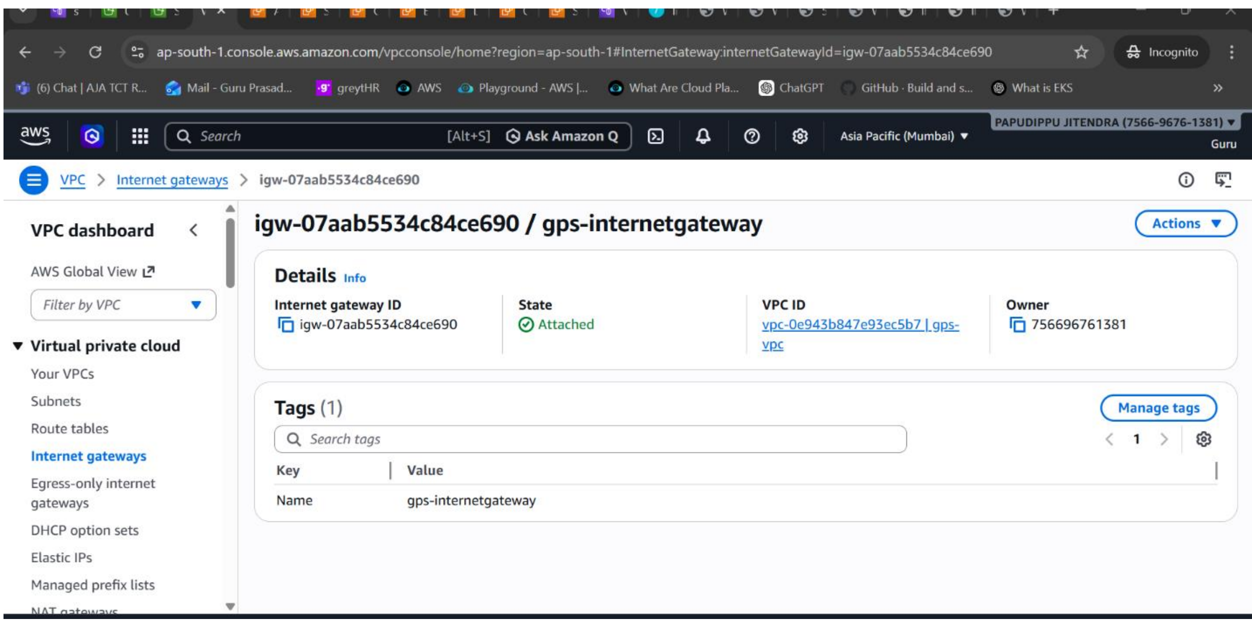
4. Attached:
   - Internet Gateway (for public access)



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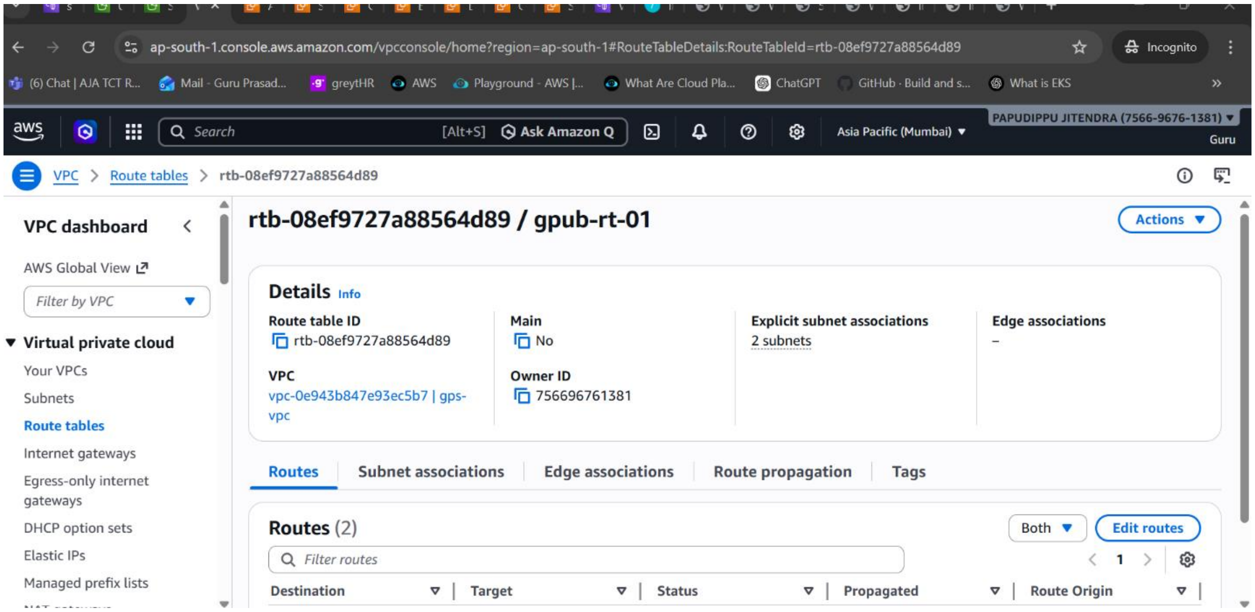
5. Configured Route Tables:
   - Public → Internet Gateway

6. 

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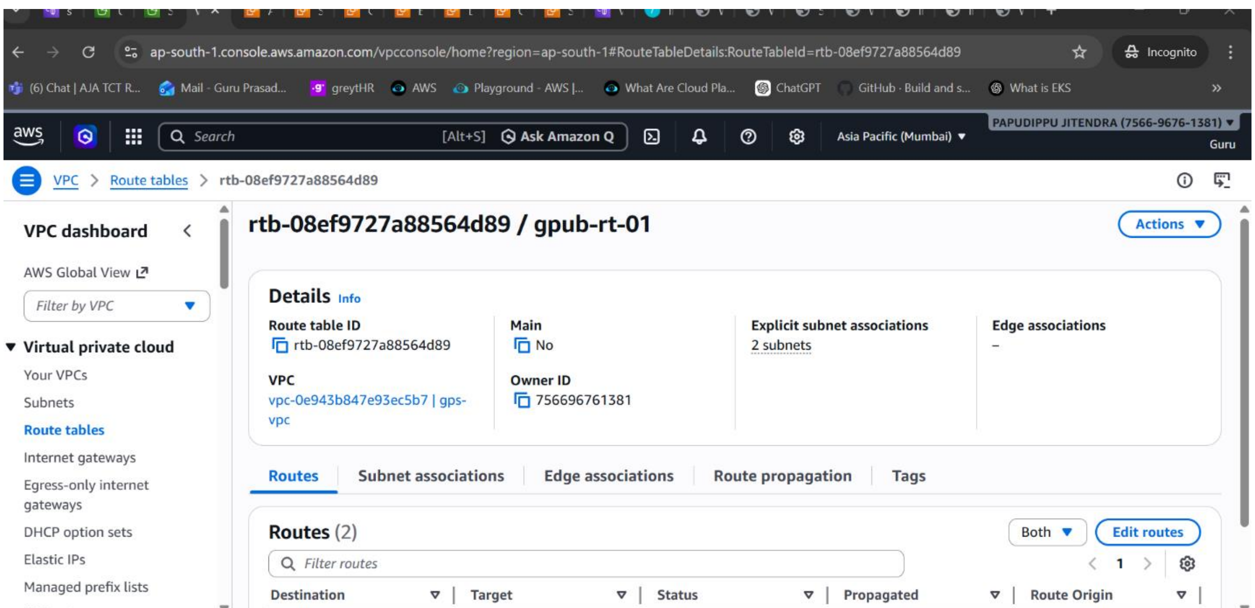
- Private → internal routing

7. 

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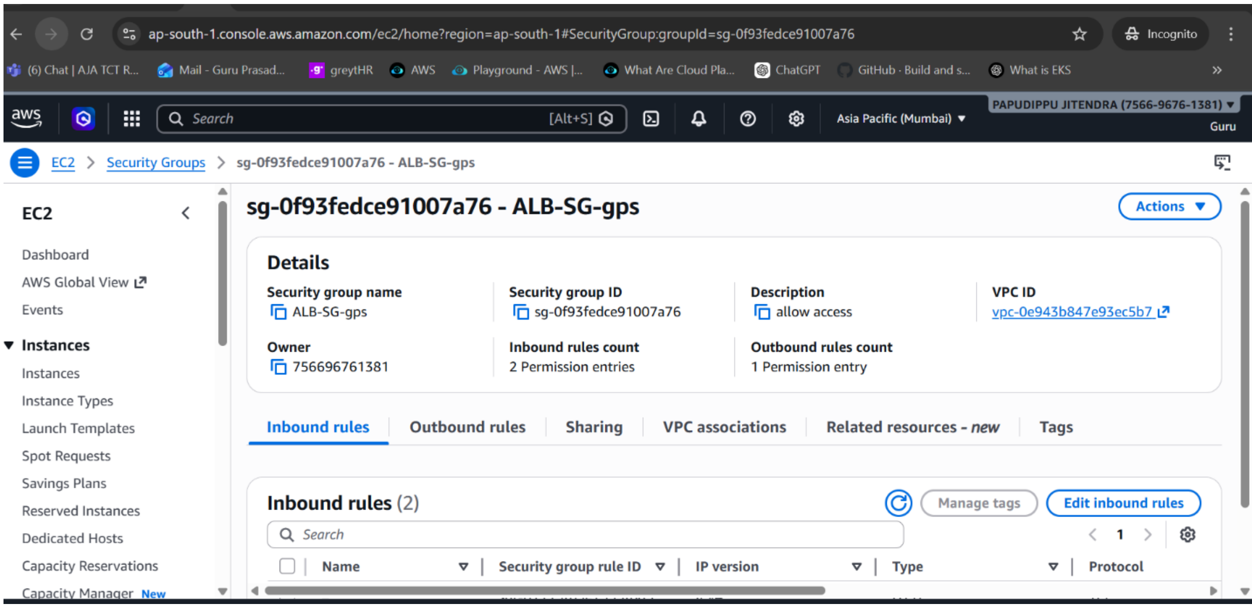
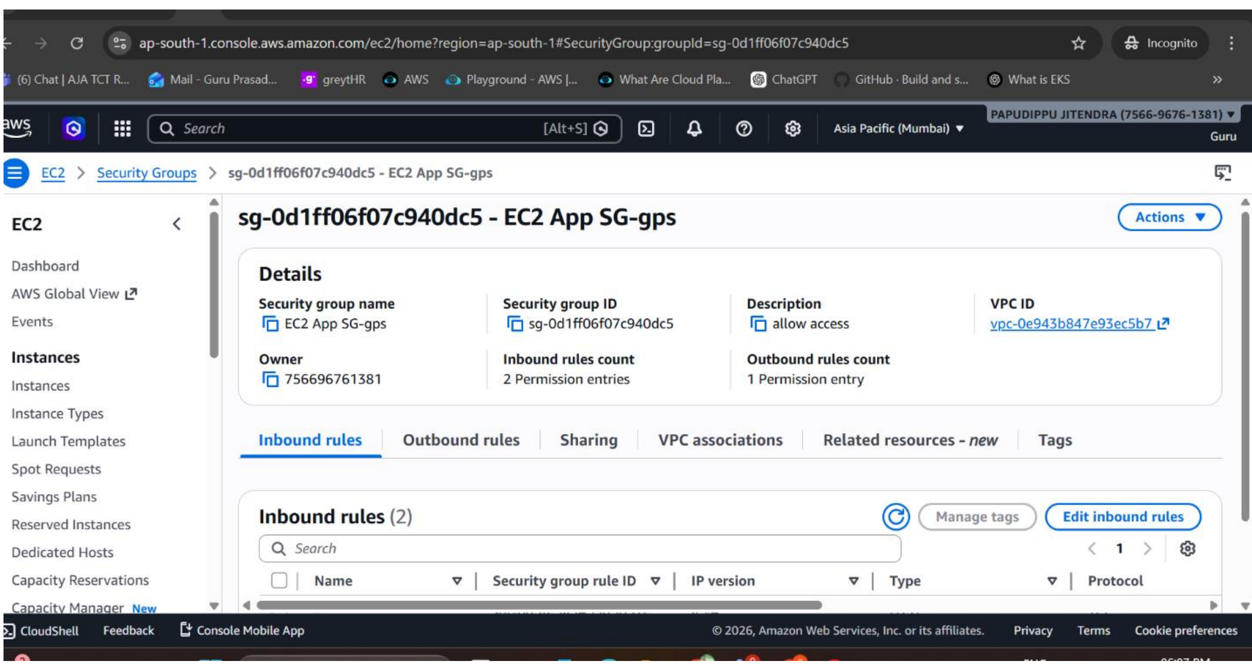
🔐 2. Security Groups Configuration

ALB Security Group:

- Allow:

    - HTTP (80) from anywhere
    - HTTPS (443) from anywhere
    - 

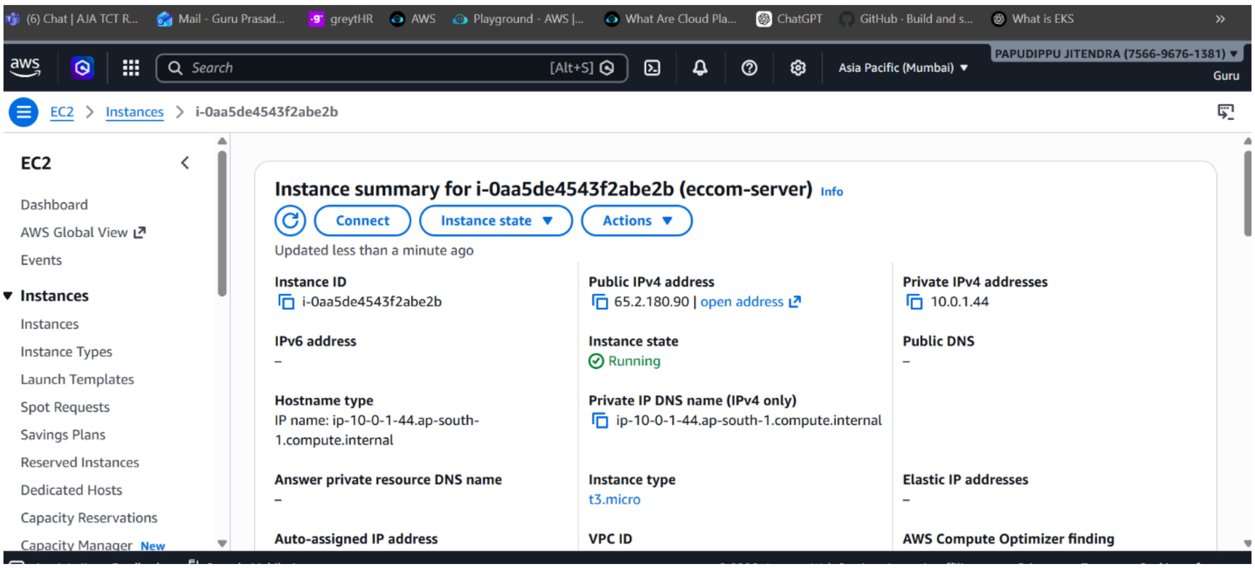
EC2 Security Group:

- Allow:
    - HTTP (80) from ALB Security Group
    - SSH (22) from Bastion Host
    - 

💻 3. EC2 & Golden AMI

Steps:

1. Launched EC2 instance



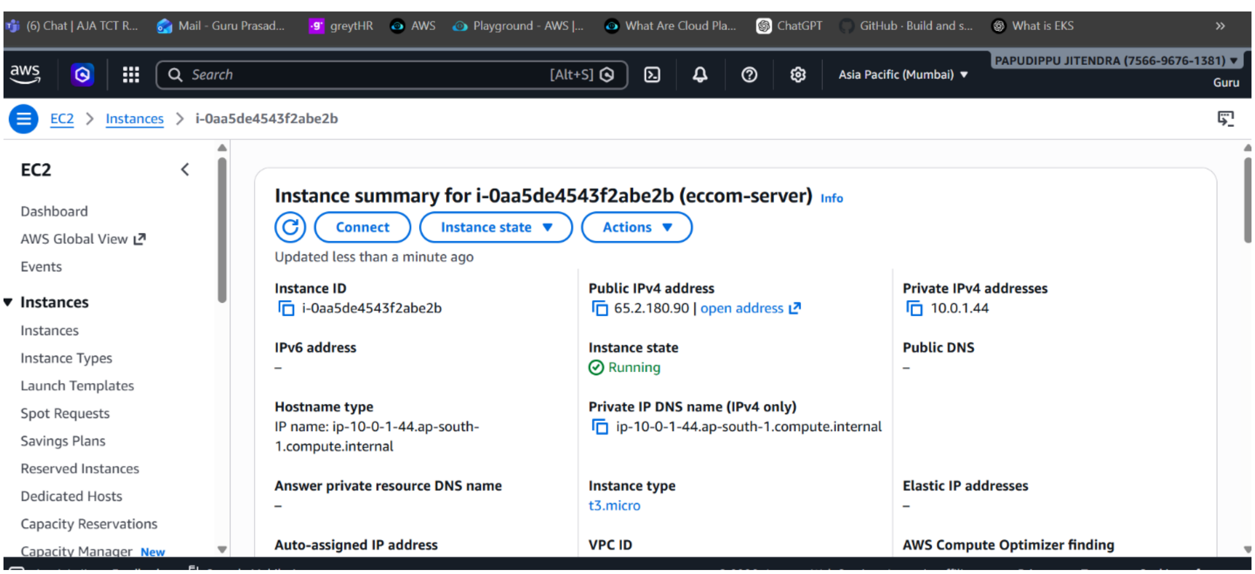
2.
3.

4. Installed web server:

sudo apt install nginx -y

Created **Golden AMI**

- Used for launching identical instances



5.

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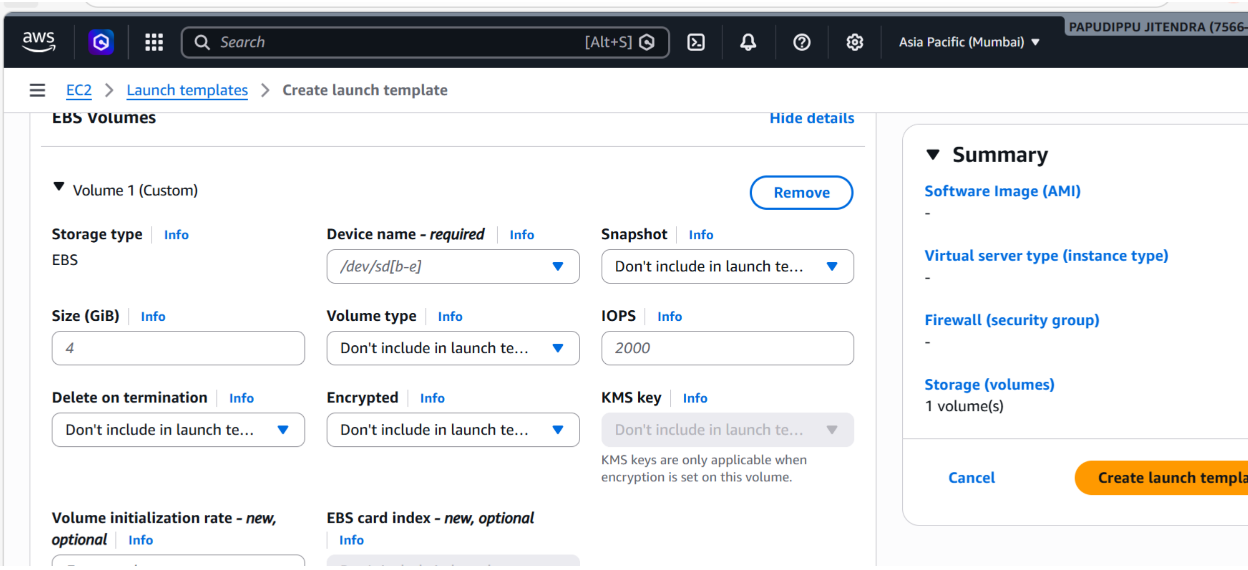
💾 4. EBS Storage

- Attached **gp3 volume (20GB)** to EC2
- Enabled:
    - Encryption using KMS
    - 

⚖️ 5. Target Group Configuration

Key Settings:

- Target Type: Instance
- Protocol: HTTP
- Port: 80
- Health Check Path:

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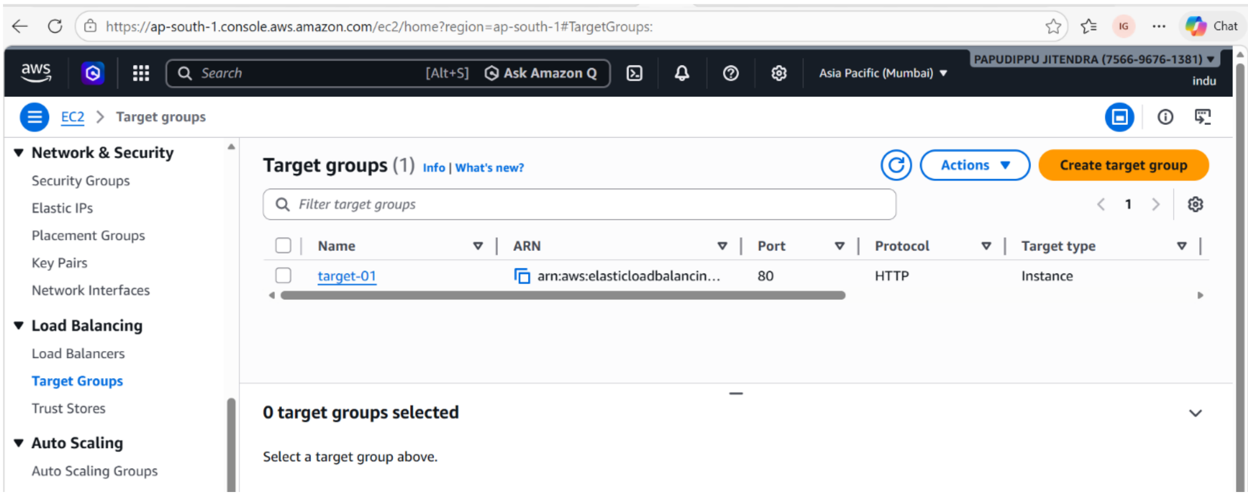
Issue Faced:

- Initially:

0 Targets Registered



- Solution:
    - Manually registered EC2 instance
    - Verified health status

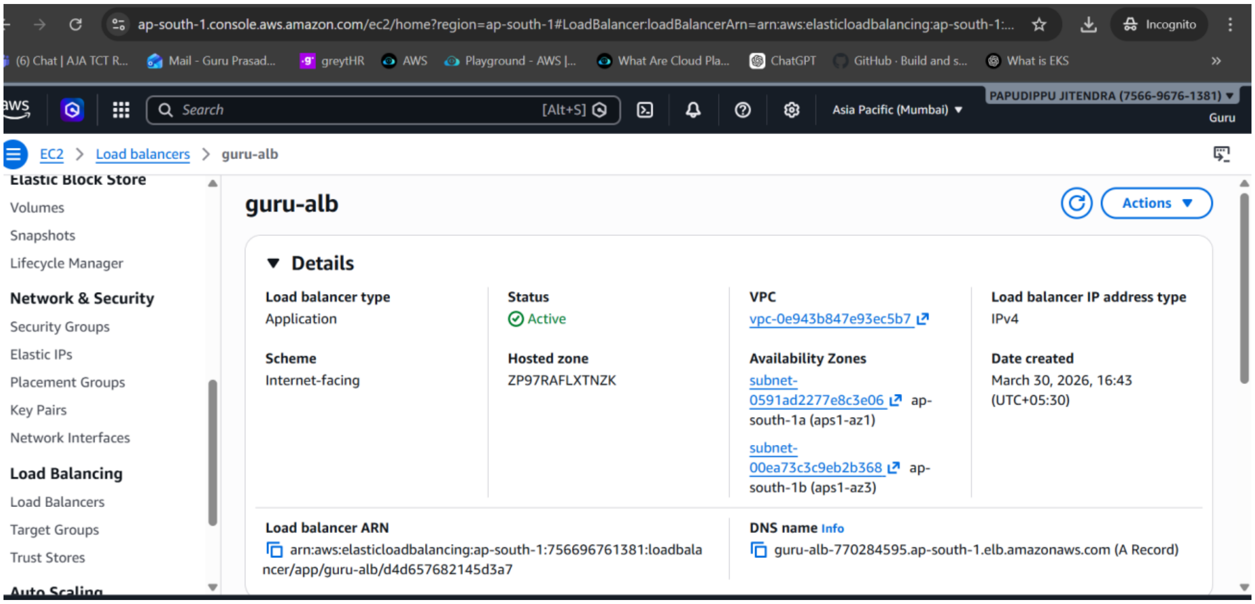
🌍 6. Application Load Balancer

Configuration:

- Type: Internet-facing
- Subnets: Public (2 AZs)
- Listener:
    - HTTP (80) → Target Group

Features Enabled:

- Cross-Zone Load Balancing
- Sticky Sessions (optional)



🔁 7. Auto Scaling Group

Configuration:

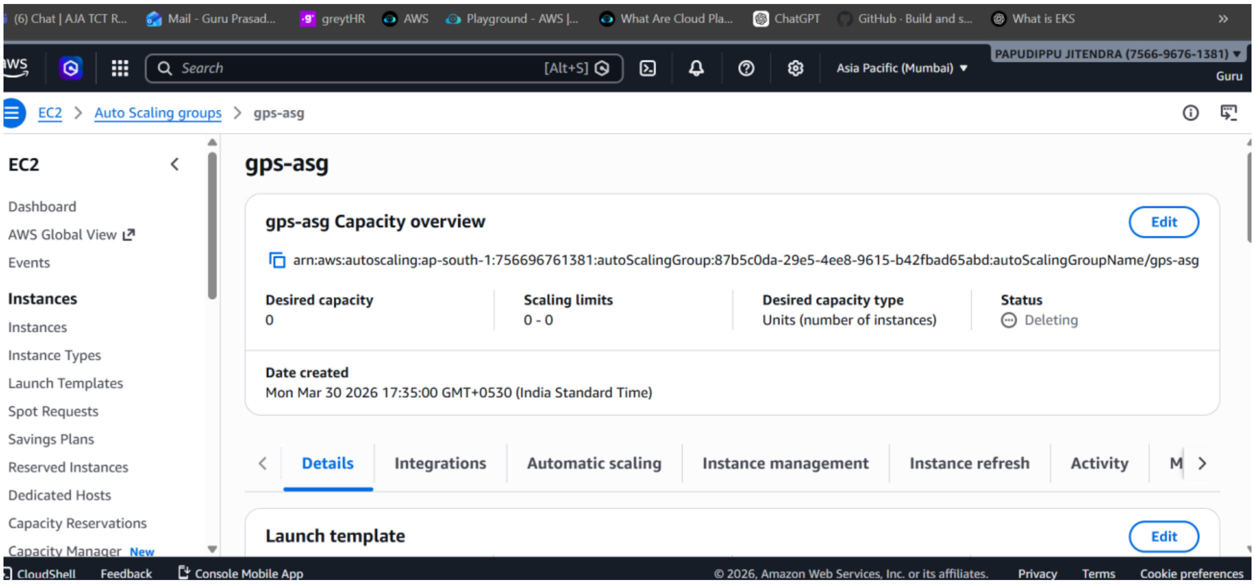
- Launch Template used (Golden AMI)
- Min: 2
- Desired: 2
- Max: 6

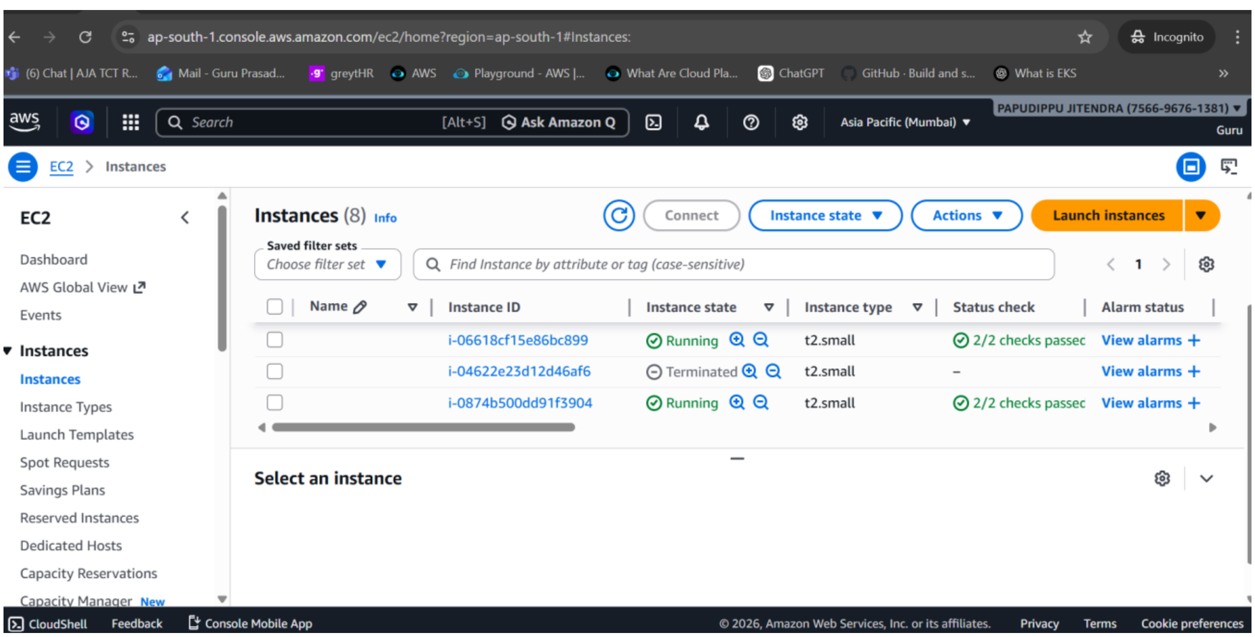
Scaling Policy:

- Scale out: CPU > 60%
- Scale in: CPU < 30%
- Cooldown: 300 seconds

Important:

- Attached Target Group to ASG
- Ensures automatic registration of instances

- 

- 

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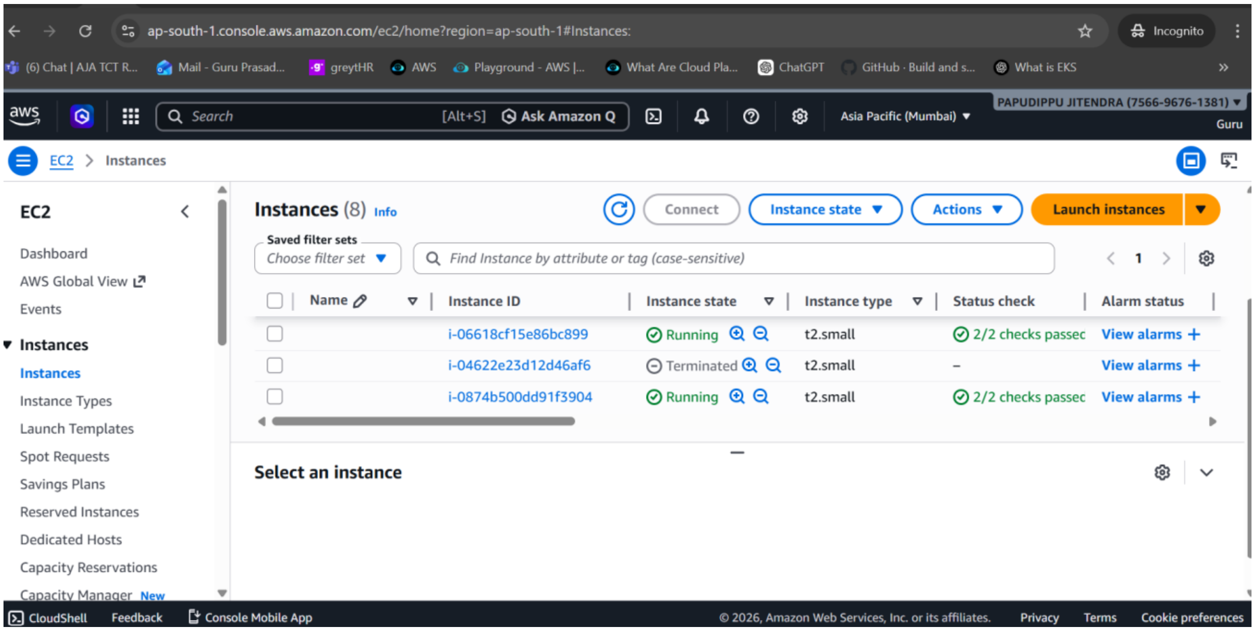
8. S3 Bucket (Static Assets)

Configuration:

- Enabled:

    - Versioning
    - Encryption (SSE-S3)

- 

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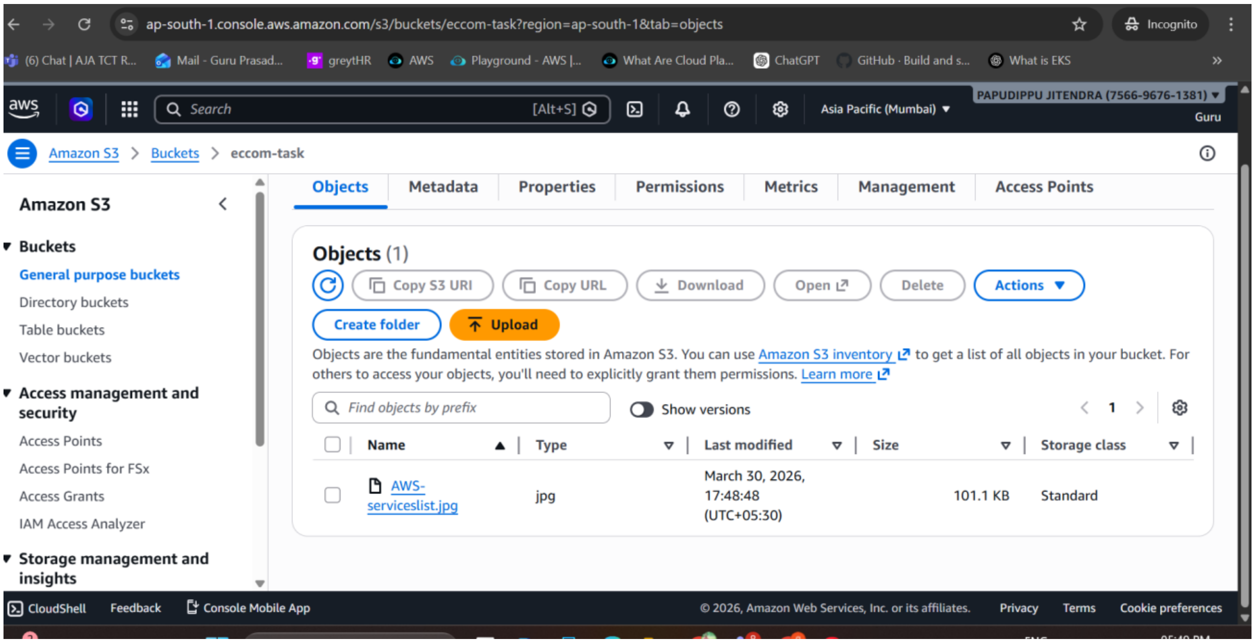
Usage:

- Stored:
    - Images
    - CSS
    - JS
    -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General purpose buckets
Directory buckets
Table buckets
Vector buckets
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Objects
Metadata
Properties
Permissions
Metrics
Management
Access Points
Objects (1)
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Actions
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Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
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